(a) Physical intervention

Load change or branch opening

Load

bus i

$P_i,\ Q_i$

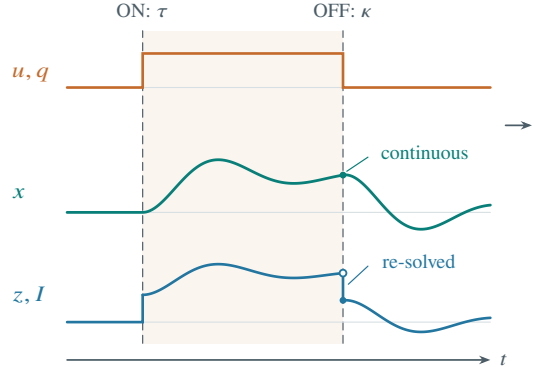
$(P_i, Q_i) \mapsto (1 + a)(P_i, Q_i)$

Line

i $\quad q_0 \mapsto q_{\text{open}}$ $\quad j$

$$\dot{x} = f_q(x, z, u)$$
$$0 = g_q(x, z, u)$$

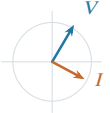
(b) State and network response



$$x(\kappa^+) = x(\kappa^-)$$
$$g_{q_0}(x(\kappa), z(\kappa^+), u_0) = 0$$

(c) Measurement formation

Eight voltage sites; mapped current terminals



Phase history η_k

$\eta_k = F_M(\eta_{k-1}, h_k)$

$h_k = H_q(x_k, z_k, u_k)$

Reported coordinates		Count
Voltage	$\lvert V\rvert, \angle V$	8 + 8
Current	$\lvert I\rvert, \angle I$	8 + 7
Frequency	$f, \dot{f}$	8 + 8

$y_k \in \mathbb{R}^{47}$ at 30 reports/s

BUS39 current angle omitted

Generic symbols and schematic trajectories; not measured or simulated records.